RP4 – Modbus Replay Detection Data Audit Report

Generated on 2026-10-23 11:52:59 | University Research Prototype

1. Uploaded Dataset Information

Dataset Source	Filename	File Size	Total Records	Capture Segments
Normal Traffic	normal_traffic_baseline.csv	Ready	150,000	6
Replay Attack Traffic	replay_attacks.csv	Ready	9,494	10

2. Traffic Ingestion & Provenance Summary

Both uploaded traffic streams have been validated at the packet/row level. Provenance metadata tracks the original source file and record index for every transaction to guarantee auditability.

3. Data Quality & Missing Value Analysis

Audit Check	Observed Count	Integrity Status	Academic Recommendation
Normal Missing Values	0	Complete	Zero missing values in mandatory MBAP headers
Replay Missing Values	0	Complete	Zero missing values in mandatory MBAP headers
Timestamp Ordering	Monotonic within capture	Verified	Discontinuities partitioned into separate capture sessions

4. Transaction Reconstruction Coverage

Transaction Metric	Count	Proportion	Research Role
Total Requests Processed	80,753	100.0%	Primary unit of detection analysis
Matched Pairs (Req-Resp)	78,620	97.36%	Valid round-trip server replies
Timeouts (Missing Resp)	2,133	2.64%	Queries eliciting no reply (indicates replay)
Orphan Responses	121	-	Server replies with no preceding query
Label Conflicts Flagged	1,047	-	Request is_attack != Response is_attack

5. Label Distribution & Conflict Analysis

Dataset Source	Class 0 (Normal)	Class 1 (Replay Attack)	Unlabeled / Unknown
Normal Traffic File	150,000	0	0
Replay Attack File	7,247	2,247	0
Combined Total	157,247	2,247	0

6. Audit Findings & Protocol Pipeline Recommendations

1. **Strict Row-Level Labeling:** The replay file contains both benign and attack packets. Labels are strictly audited at row level and never inferred from filenames.

2. **Transaction-Level Context Scoping:** Transactions are paired using 8-tuple (capture_id, client_ip, server_ip, client_port, server_port, TID, Unit ID, Function Code) rather than TID alone, preventing identifier collision.

3. **Temporal Leakage Elimination:** Data splitting employs independent capture files and chronological blocking with a 300-second separation gap to guarantee test isolation.

4. **Verification Status:** Data audit verified 100% integrity across all required protocol fields. The datasets are certified ready for feature extraction and model training.